BACHELOR OF COMPUTER APPLICATIONS (BCA)

SEMESTER VI - END SEMESTER EXAMINATION

May 2024

Paper Code: BCA-603
Subject: Software Engineering

Time Allowed: 3 Hours | Maximum Marks: 75

General Instructions:

1. Attempt all sections. Section A is compulsory; internal choice is given in Sections B and C.
2. Write answers in legible handwriting and number them exactly as in the question paper.
3. Draw neat, labelled diagrams wherever required.

SECTION A (5 x 3 = 15 Marks)

Answer all five questions. Each question carries 3 marks.

1. Define software engineering. [3]

2. What is a software requirement specification (SRS)? [3]

3. Differentiate between verification and validation. [3]

4. What is a prototype? [3]

5. Define software maintenance. [3]

SECTION B (4 x 10 = 40 Marks)

Answer any four questions. Each question carries 10 marks.

1. Explain the waterfall model with its advantages and limitations. [10]

2. Explain functional and non-functional requirements with examples. [10]

3. Describe black-box and white-box testing techniques. [10]

4. Explain the COCOMO estimation model. [10]

SECTION C (2 x 10 = 20 Marks)

Answer any two questions. Each question carries 10 marks.

1. Explain the Spiral model and compare it with the incremental model. [10]

2. Discuss the phases of the software development life cycle and the deliverables of each phase. [10]

*** *END OF PAPER* ***